

Chapter 12 System Analysis and Design

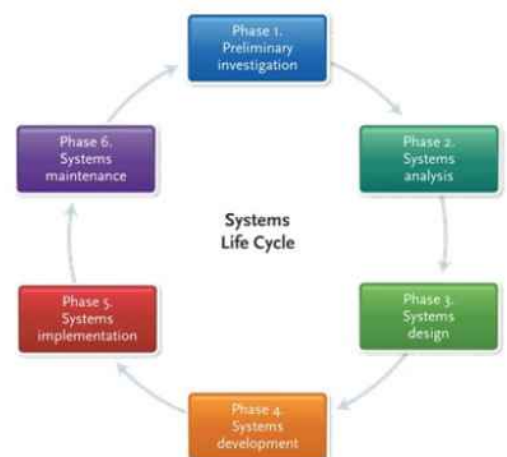
Introduction

- Most people in an organization are involved with an information system of some kind.
- For the organization to create and use the system requires thought and effort.
- Need to be aware of the relationship of an organization's chart to its managerial structure.

System Analysis and Design

A system is defined as a collection of activities and elements organized to accomplish a goal

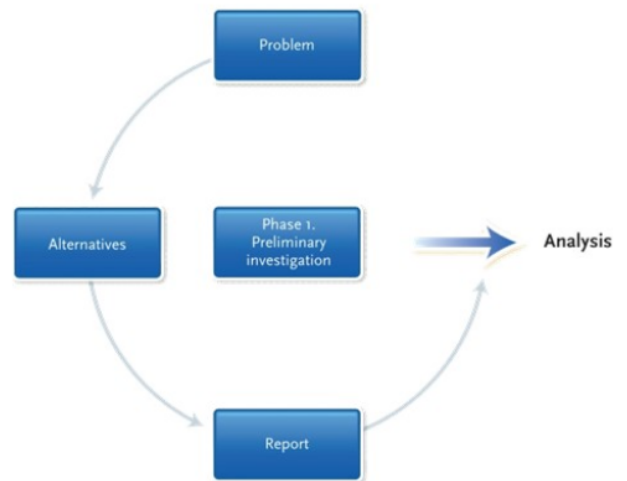
- System Analysis and Design
 - Six-phase problem-solving procedure for examining and improving an information system



Phase 1: Preliminary Investigation

The preliminary investigation determines the need for a new information system

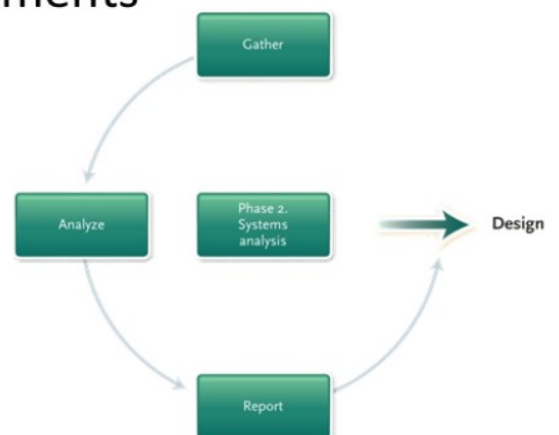
1. Define the problem
2. Suggest alternatives
3. Prepare report



Phase 2: Systems Analysis

Data is collected about the present system and then analyzed to determine the new requirements

1. Gather data
2. Analyze data
3. Create summary



Analyzing the Data in Phase 2

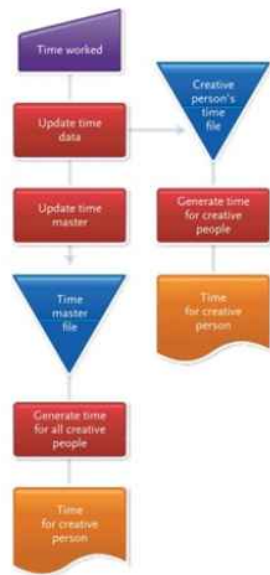
- Top-down analysis method is used to identify the top-level components of a complex system and each component is broken down into small components making analysis easier
 - Grid charts show the relationship between input and output documents
 - System flowcharts show the flow of input data to processing and finally to output or distribution of information
 - Data flow diagrams show the data or information flow within an information system
 - Automated design tools
 - Computer-aided software engineering tools (CASE)
 - Documenting
 - Systems Analyst Report

Data Analysis Tools – Grid Chart

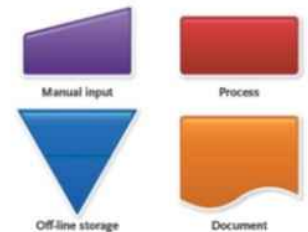
Show the relationship between input and output documents

Forms (input)	Reports (output)		
	Client billing	Personnel expense	Support cost
Time sheet	✓	✓	
Telephone log	✓		✓
Travel log	✓		✓

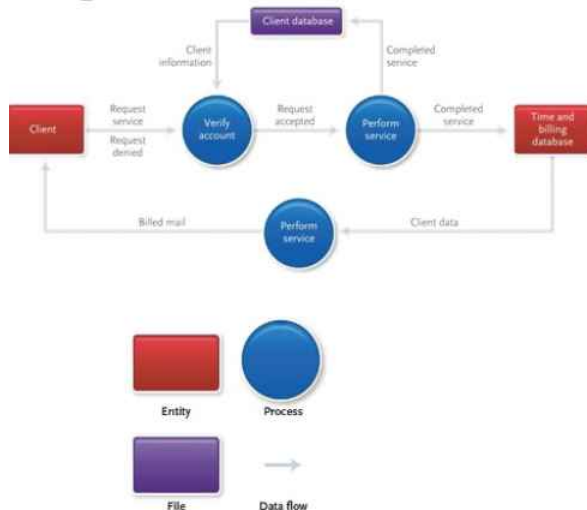
Data Analysis Tools – System Flowchart



- System flowcharts show the flow of input data to processing and finally to output or distribution of information.
- System flowchart is to the left
- System flowchart symbols are to the right



Data Analysis Tools – Data Flow Diagram

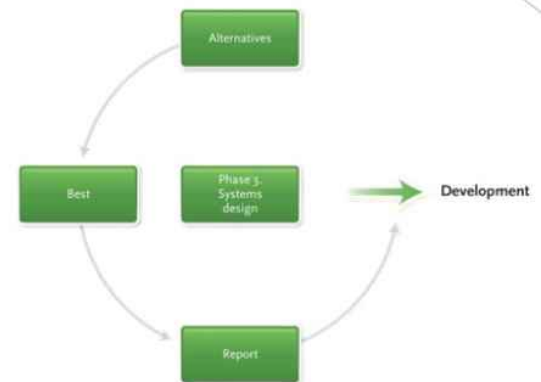


- Show the data or information flow within an information system
- Data is traced from its origin through processing, storage, and output
- Top diagram shows data flow
- Bottom diagram shows data flow symbols

Phase 3: Systems Design

Three tasks

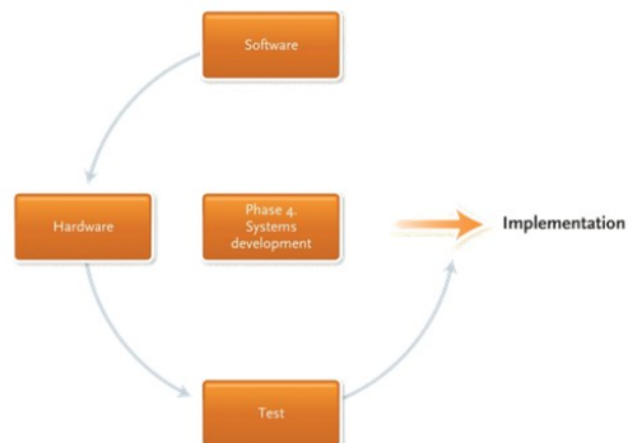
- Define the alternatives
- Select the best system
- Write a systems design report
- Evaluate systems according to:
 - Economic feasibility asks is a new system be economical?
 - Technical feasibility asks is it technically possible?
 - Operational feasibility asks how it will be received by all users



Phase 4: Systems Development

Three steps

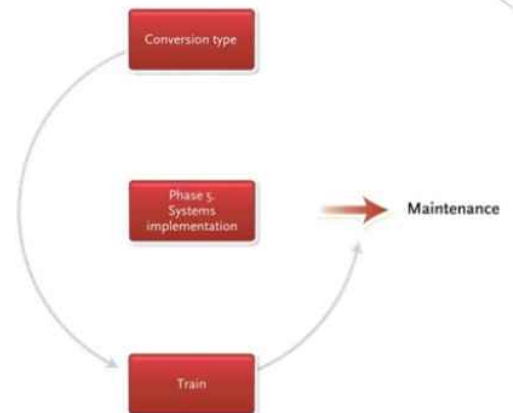
1. Acquire the software
2. Acquire the hardware
3. Test the new system



Phase 5: Systems Implementation

Types of conversion

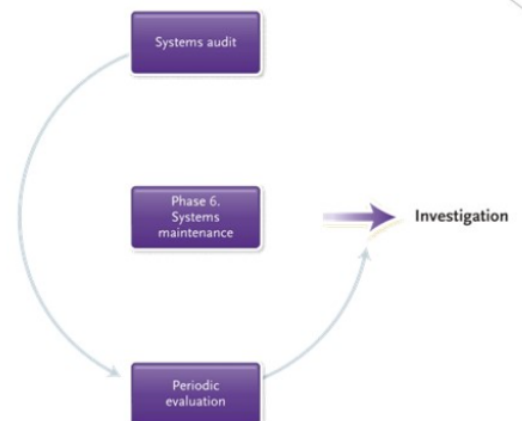
- Direct – out with the old and in with the new
- Parallel – old and new systems operate side by side until new system is reliable
- Pilot – new system is tried by one section of the organization, then another, etc.
- Phased – new system is introduced a little at a time



Phase 6: Systems Maintenance

Systems maintenance is an ongoing activity

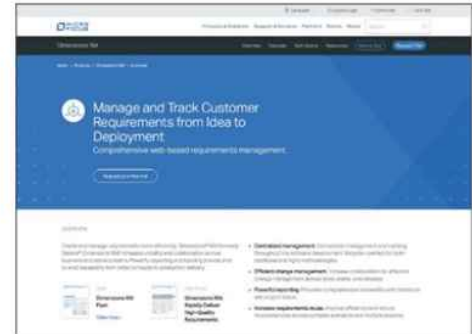
- Two Parts
 - Systems audit - system's performance is compared to the original design specs to determine productivity
 - Periodic evaluation – all system's should be evaluated time to time



Prototyping and Rapid Applications Development

Alternatives to the systems life cycle

- Prototyping - building a model of the new system for trial
- Rapid applications development (RAD) – costly but development is short and quality is better



Careers in IT

- A systems analyst plans and designs new systems, following the systems life cycle
- Requires a Bachelor's degree in Computer Science or Information Systems
- Technical experience
- Can expect to earn an annual salary of \$53,000 to \$77,000



A Look to the Future ~ The Challenge of Keeping Pace

- To stay competitive with today's fast business pace, new technologies must be incorporated
- Increased use of RAD and prototyping
- Increased use of outside consulting

